

LANSERIA, South Africa

WMO: 682635

Lat: 25.939S

Lon: 27.926E

Elev: 4517

StdP: 12.45

Time Zone: 2.00 (E02)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.4	37.5	10.0	10.8	60.4	13.6	13.0	60.6	21.7	59.0	18.9	62.1	2.9	240	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.5	91.1	61.4	88.2	61.3	86.4	61.5	68.4	78.7	67.5	77.7	66.9	77.2	12.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	114.4	70.1	64.7	108.6	69.8	64.3	106.8	69.7	35.7	79.4	34.9	77.9	34.3	76.7	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.8	15.5	DB	30.2	94.1	2.5	2.2	28.4	95.6	26.9	96.9	25.5	98.1	23.6	99.7	(4)
(5)			WB	26.3	70.6	1.7	1.1	25.1	71.3	24.1	72.0	23.1	72.6	21.9	73.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	72.3	72.3	69.9	64.0	58.8	53.7	53.1	58.4	65.9	69.4	70.7	72.2	(6)	
	DBStd	8.14	3.44	3.00	3.27	3.93	3.63	3.60	3.46	4.93	5.01	5.15	4.84	4.33	(7)	
	HDD50	28	0	0	0	0	1	10	13	4	0	0	0	0	(8)	
	HDD65	1266	2	1	4	64	192	340	370	209	48	21	10	5	(9)	
	CDD50	5508	691	625	617	419	275	120	108	264	476	602	622	689	(10)	
	CDD65	1273	228	206	156	33	1	0	0	5	74	158	183	229	(11)	
	CDH74	11521	1527	1417	1169	388	121	11	12	232	1202	1789	1770	1883	(12)	
	CDH80	3499	457	390	291	50	6	0	0	18	348	651	627	661	(13)	
(14) Wind	WSAvg	6.3	6.6	6.3	5.9	5.3	4.7	5.0	5.1	6.2	7.1	8.0	8.2	7.3	(14)	
(15) Precipitation	PrecAvg	23.7	4.4	3.7	3.4	1.3	0.6	0.1	0.0	0.1	0.5	2.2	3.3	4.1	(15)	
	PrecMax	39.8	9.6	9.4	12.2	4.2	3.2	0.6	0.3	0.8	1.8	6.2	6.9	7.0	(16)	
	PrecMin	13.1	0.9	0.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.3	1.6	(17)	
	PrecStd	6.8	2.2	2.5	2.8	1.1	0.8	0.2	0.1	0.2	0.6	1.6	1.6	1.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	91.2	89.3	84.0	80.7	75.5	75.3	82.5	89.6	93.2	93.4	93.0	(19)	
		MCWB	62.8	63.3	60.8	61.2	54.7	51.5	49.8	54.7	57.1	60.3	59.6	63.3	(20)	
	2%	DB	87.9	87.6	86.1	80.9	77.2	73.0	72.0	79.1	87.4	89.7	89.4	89.6	(21)	
		MCWB	64.0	63.7	62.0	59.1	53.8	50.1	48.8	53.6	56.4	58.5	60.7	63.9	(22)	
	5%	DB	84.6	84.4	83.9	78.6	75.2	70.1	69.9	76.9	84.4	87.4	86.4	86.4	(23)	
		MCWB	64.9	64.4	61.4	59.0	53.5	49.2	48.7	52.8	55.5	58.7	61.2	63.8	(24)	
	10%	DB	82.4	82.4	80.8	75.4	73.1	67.9	67.6	73.6	82.1	84.0	84.0	84.1	(25)	
		MCWB	65.3	64.7	61.9	58.3	53.4	48.9	48.3	50.9	54.8	58.7	61.2	63.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.0	69.2	68.2	64.8	61.4	55.1	53.8	57.4	60.9	64.9	67.1	69.5	(27)	
		MCDB	81.1	80.0	77.1	77.6	70.4	66.8	66.0	77.5	79.3	79.2	80.4	80.1	(28)	
	2%	WB	68.6	68.0	66.8	63.3	58.5	53.4	51.9	55.8	59.5	63.6	65.6	67.9	(29)	
		MCDB	78.6	78.6	75.6	74.8	69.0	65.5	65.0	74.6	77.5	78.1	79.5	79.7	(30)	
	5%	WB	67.6	67.2	65.6	61.9	56.7	51.8	50.4	54.5	58.3	62.2	64.7	66.9	(31)	
		MCDB	76.9	77.8	74.5	72.4	69.1	64.4	64.5	72.0	76.0	76.8	77.8	78.5	(32)	
	10%	WB	67.0	66.4	64.5	60.4	55.3	50.5	49.3	53.0	57.0	61.3	63.8	65.9	(33)	
		MCDB	76.1	77.0	73.4	70.1	67.7	63.2	63.9	69.2	74.3	75.8	76.5	77.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	19.9	20.9	22.4	27.3	28.0	28.0	28.5	28.2	25.6	22.9	20.5	(35)	
		MCDBR	23.9	24.2	26.2	27.1	30.8	31.8	32.4	31.8	32.3	30.0	27.5	25.5	(36)	
	5% WB	MCWBR	8.1	7.8	8.6	9.2	12.1	14.1	14.2	13.3	12.1	9.9	9.1	8.0	(37)	
		MCWBR	18.9	20.4	19.5	21.1	24.7	26.3	26.5	27.7	26.0	24.5	22.8	20.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.398	0.401	0.390	0.341	0.302	0.284	0.286	0.343	0.386	0.419	0.378	0.385	(40)	
		taud	2.312	2.301	2.312	2.418	2.497	2.521	2.501	2.339	2.236	2.179	2.338	2.339	(41)	
	Ebn at Noon		300	294	288	288	287	287	291	283	284	285	304	305	(42)	
		Edh at Noon	44	43	41	34	28	26	28	36	43	49	43	43	(43)	
(44) All-Sky Solar Radiation	RadAvg		2106	1992	1767	1514	1387	1258	1366	1611	1919	2059	2143	2157	(44)	
	RadStd		221	180	163	134	69	62	35	74	86	140	195	130	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	-326	+421	+157

Nomenclature: See separate page